

Evaluation Physicochemical Drivers of River Flow

Using Weighted Regression

Note: This version of the report has been condensed for readability. A more comprehensive analysis, including full R code and output, is provided in the accompanying code report.

Abstract

This project seeks to investigate if physicochemical properties of the Brisbane River, including pH, dissolved oxygen, salinity, chlorophyll, and turbidity, are associated with average water speed. The data provided by the Queensland Government Open Data Portal contains over 20,000 observations collected at 30-minute intervals of the river. After addressing missing values and multicollinearity, a multiple linear regression model was fit and refined using stepwise selection under the Bayesian Information Criterion (BIC). Diagnostic testing revealed heteroscedasticity, which was then managed by using Weighted Least Squares, removing extreme outliers to improve the model's stability. The final model identified all five predictors as statistically significant ($p < 0.01$). Dissolved oxygen and salinity were associated with lower water speeds, while pH and turbidity were associated with higher speeds. Overall, the model explained a modest proportion of the variability in water speed, with an adjusted $R^2 \approx 0.174$. Evaluation on a held-out test set demonstrated that the model maintained consistent performance on unseen data (RMSE ≈ 12.02 cm/s, MAE ≈ 8.98 cm/s), suggesting that its moderate explanatory power reflects underlying variability in river dynamics rather than substantial overfitting.

Introduction

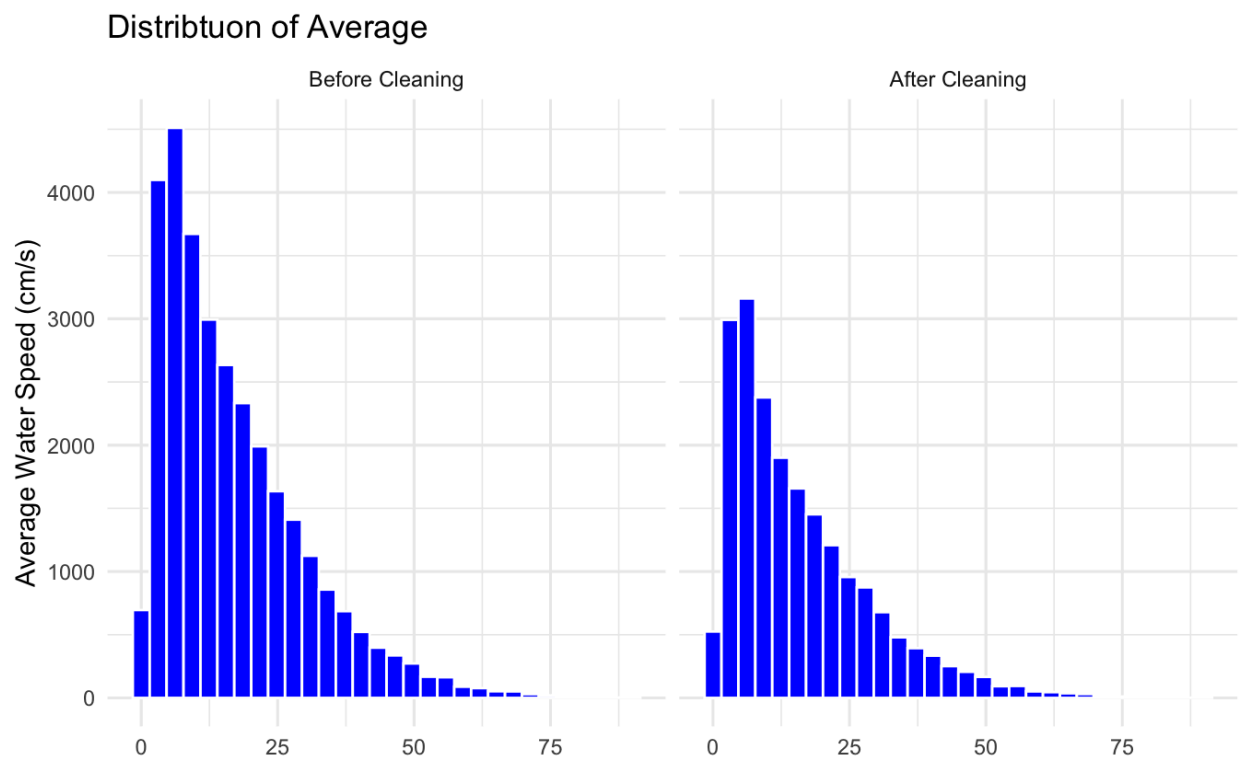
This project examines a dataset provided by the Queensland Government Open Data Portal containing measurements of water physicochemical parameters of the Brisbane River in Australia (Fedorov). The data includes timestamps, record numbers, and various characteristics of the river during a given instance. Using this data, we aim to test whether or not the average water speed is influenced by these river characteristics. Understanding water speed is crucial because it affects numerous things: river habitats and aquatic biodiversity, agriculture,

hydroelectric power generation, erosion, flood risk, management, and many others. Monitoring these speeds is tantamount to ensuring risk management and environmental protection (Fain).

Section 1: Descriptive Statistics and Checking Assumptions

We begin our analysis by first examining some characteristics of the dataset. Descriptive statistics are helpful in arming us with useful data points as well as allowing us to detect potential outliers whether that be highly correlated variables or null values (NA). We will ignore columns Timestamp column as it may over complicate our analysis and the various [quality] columns due to their long stretch of null values.

This dataset contains over 30,000 observations of the Brisbane River, capturing variation in the water's physicochemical parameters across 30-minute intervals. Before cleaning, many variables contained a substantial amount of missing values, most notably temperature (5164 missing), dissolved oxygen (4300 missing), salinity (3958 missing), and turbidity (2000 missing). After removing these rows to ensure we only have complete observations, the dataset retained the majority of its observations while dropping potential inconsistencies.



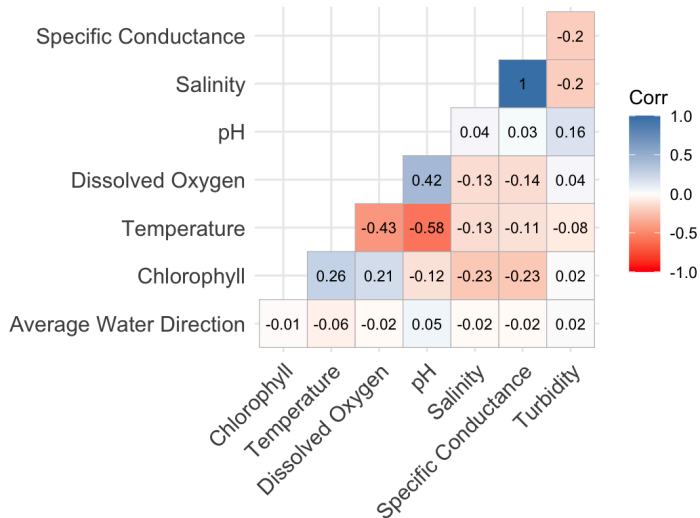
Comparing the summary statistics before and after cleaning the data, some key trends slightly shifted, reflecting the removal of those null/extreme values. For instance, the mean average water speed decreased from ≈ 16.80 to ≈ 16.01 , and the median from ≈ 13.39 to ≈ 12.33 ; the range remained wide, indicating some persistent extreme values. With regards to the other average water direction and the physicochemical parameters, they also exhibited minor changes.

Overall, the removal of these missing values slightly adjusted the descriptive statistics, retaining variability and the general patterns of the dataset. This version provides us with a more reliable starting point for further analysis.

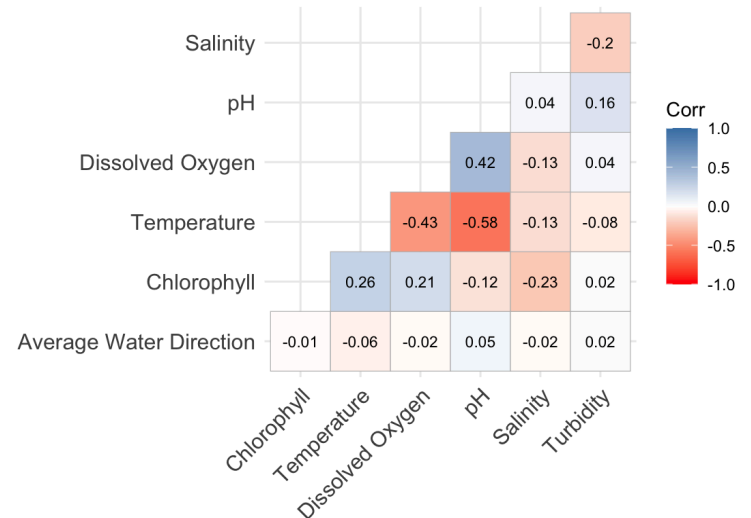
Section 2: Model Preparation and Fitting

We first examined correlation structure to avoid multicollinearity. Salinity and Specific Conductance were nearly perfectly correlated (≈ 0.999), so Specific Conductance was removed. Remaining predictors all fell below common correlation thresholds.

Correlation Matrix of Physicochemical Predictors



Correlation Matrix of Physicochemical Predictors (Omitted Conductance)



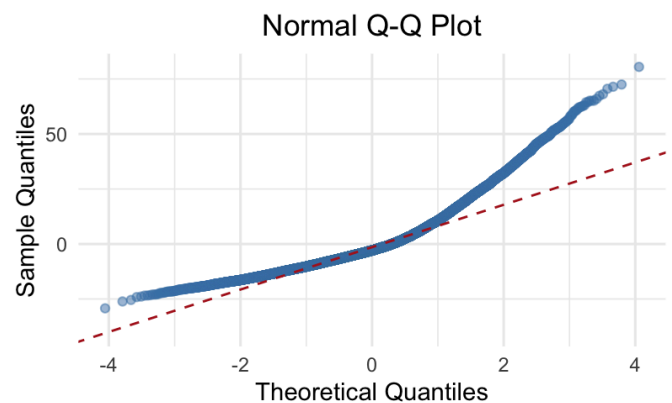
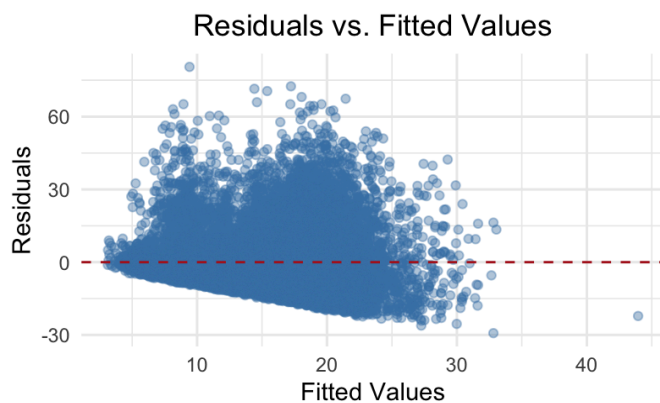
We fitted a multiple linear regression model using average water speed as the response and the remaining physicochemical variables as predictors. Temperature and Average Water Direction were removed due to non-significant p-values, resulting in a final model including Chlorophyll, Dissolved Oxygen, pH, Salinity, and Turbidity. Stepwise selection using BIC selected the same model, supporting its adequacy.

The adjusted R^2 of 0.137 indicates that our model explains a moderate portion of variation in average water speed. Much of the remaining variability is likely driven by environmental factors not captured in this dataset. Regardless, our predictors' statistical significance do prove that they truly have an effect on determining average water speed, even if the model doesn't explain most of the variation. Rivers are inherently dynamic, as their flow can be strongly affected by a variety of external forces and conditions (Trent and Brown).

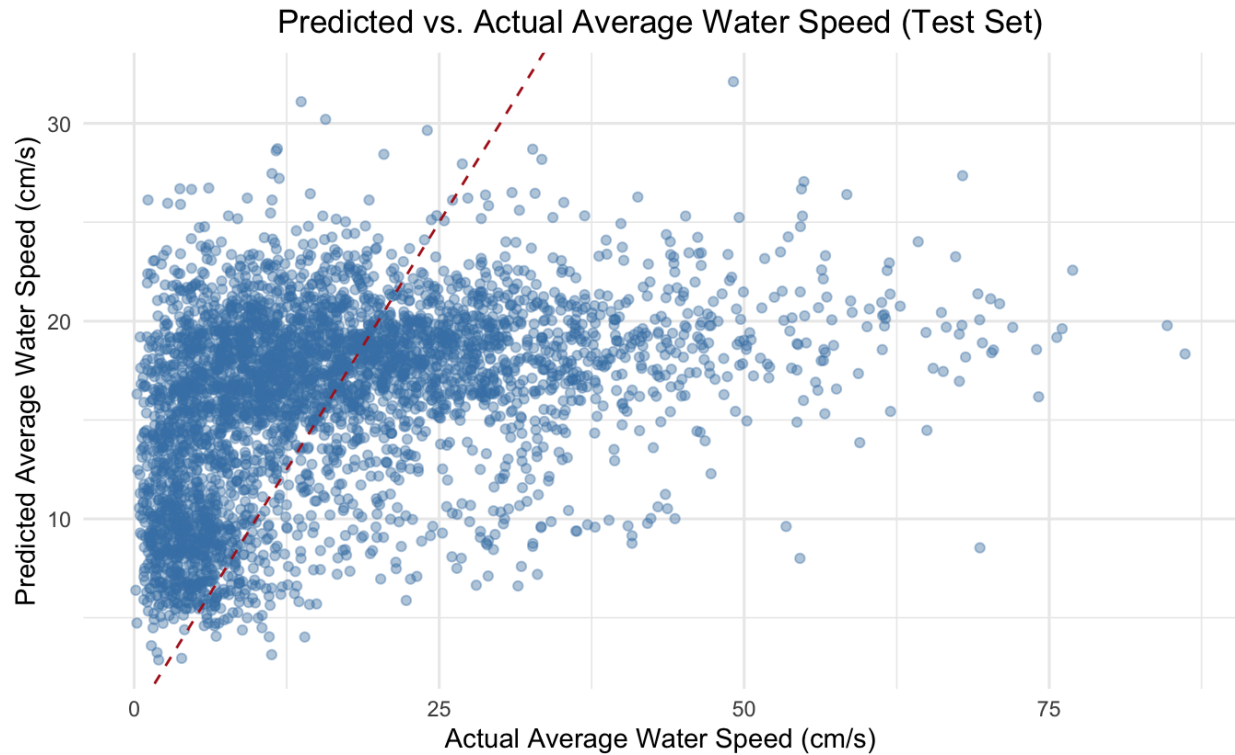
Furthermore, the Variance Inflation Factor (VIF) values for all predictors are well below the common threshold of 5, indicating that multicollinearity is not a concern of this model.

Section 3: Diagnostics

Residual-versus-fitted plots and a Breusch–Pagan test indicated strong heteroscedasticity. Q–Q and component-plus-residual plots suggested minor nonlinearity but not enough to justify transforming the response; a Box–Cox test confirmed an optimal lambda of 1.

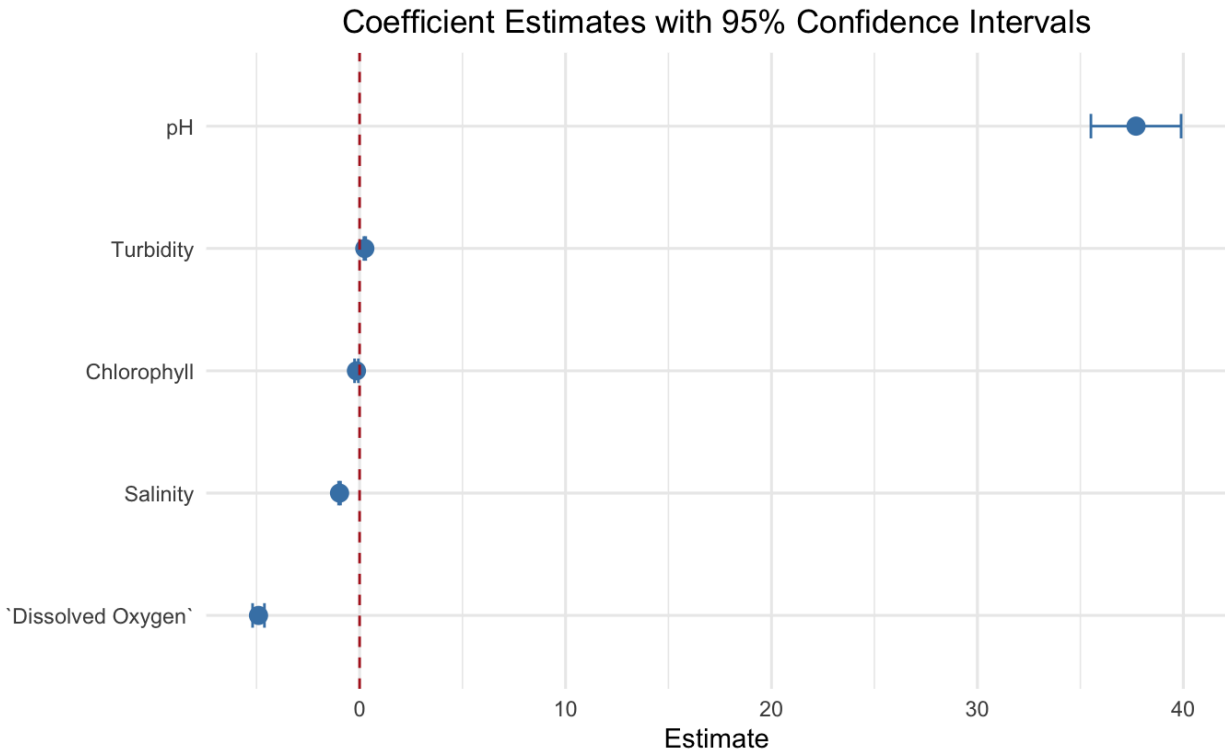


Weighted Least Squares (WLS) was applied to address non-constant variance, improving the BP test result and reducing heteroscedasticity. Ten observations with extreme water speeds were flagged as outliers and removed. After refitting the WLS model, heteroscedasticity was further reduced, though not eliminated entirely.



Section 5: Analysis

The final weighted least squares model depicts how Chlorophyll, Dissolved Oxygen, pH, Salinity, and Turbidity influence the average water speed of the Brisbane River. Within the parameter estimates, all coefficients are statistically significant at the 0.01 level, indicating well-supported points. Notably, dissolved oxygen has a strong negative effect (Estimate ≈ -4.91), suggesting that higher oxygen levels are correlated with lower water speeds. pH displays a very strong, positive effect (Estimate ≈ 37.73), implying higher pH associates with faster water speeds. Salinity has a moderately negative effect (Estimate ≈ -0.97), while Turbidity (Estimate ≈ 0.25) and Chlorophyll (Estimate ≈ -0.15) produced smaller but still significant effects. The model explains roughly 17.4% of variability in water speed, suggesting that other environmental factors not included within the dataset likely contribute to variation.



Section 6: Limitations and Future Work

While the final model provides statistically significant and interpretable relationships between physiochemical parameters and average water speed, several limitations should be acknowledged. First, the model explains only a modest portion of the variability in water speed (Adjusted $R^2 \approx 0.174$), suggesting that important drivers of river flow are not captured in this dataset. According to the research (Less et al. 1181), hydrological factors such as rainfall, tidal cycles, and upstream discharge volume are known to strongly influence river velocity. However, these were unavailable in the source data, and their absence likely accounts for much of the unexplained variance.

Second, the Timestamp variable was excluded from modeling to simplify the analysis, meaning temporal structure such as diurnal tidal patterns or seasonal variation, was not directly incorporated. Given that river flow is inherently time-dependent, future work could explore this structure directly, either through time-series methods that account for autocorrelation between

sequential 30-minute readings, or through mixed-effects models that treat time-of-day or season as structured random effects.

Finally, the train/test validation performed did give us confirmation that the data generalizes well with unseen data, but this could be further extended using k-fold cross-validation to give us a greater range of estimates across many partitions to better defend our findings.

Conclusion

This analysis provides a comprehensive assessment of how key physicochemical parameters influence the average water speed of the Brisbane River. Using multiple linear regression and rigidly considering model assumptions, we identified Chlorophyll, Dissolved Oxygen, pH, Salinity, and Turbidity as significant predictors of water speed. The Weighted Least Squares approach improved the reliability of coefficient estimates by reducing the influence of heteroscedasticity, and removal of extreme outliers further strengthened model stability. While the final model explains approximately 17.4% of the variability in water speed, the statistical significance of all predictors demonstrates that these environmental factors meaningfully contribute to river flow dynamics. Notably higher Dissolved Oxygen and Salinity are associated with slower flows, whereas higher pH and Turbidity relate to faster speeds, highlighting the various complex interactions of physicochemical properties in shaping river behavior.

Overall, this model offers a robust and interpretable scheme for understanding river flow, emphasizing that while rivers are inherently dynamic and influenced by a multifaceted range of external factors, careful monitoring of these key parameters can provide valuable insights for environmental management, risk mitigation, and ecological preservation.

Works Cited

1. Fain, Jeremy. "Understanding Surface Water Speed in Rivers: Why It Matters and How It's Measured - Blue Water Intelligence (BWI)." *Blue Water Intelligence (BWI)*, 8 Sept. 2025, <https://bwi.earth/understanding-surface-water-speed-in-rivers-why-it-matters-and-how-its-measured/>.
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